Data Reduction via Query-Aware Community-based Subset Selection

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Abstract

Large multimedia datasets often contain redundant elements that can be removed while preserving the information content of the original collection. This project investigates subset selection strategies for data reduction, comparing exact optimization, IndepDF, Shapley-value-based selection, and a proposed community-based selection approach. The proposed method combines query-based relevance estimation with community detection over query co-occurrence patterns and optionally applies Maximum Marginal Relevance (MMR) to promote embedding-space diversity. The methods are evaluated in terms of query-answer preservation, embedding diversity, and computational efficiency. Experimental results show that, on the full real-world and synthetic datasets, community-based selection consistently improves the data forgetting objective over IndepDF, particularly under small selection budgets, while introducing additional computational overhead. Results on the restricted candidate pool further show that the effectiveness of community-based selection depends on the relationship between the selection budget and the detected community structure. The MMR analysis shows that the relevance-diversity trade-off is dataset- and budget-dependent, suggesting that the balance between relevance and diversity should be tuned according to the characteristics of the workload.

Keywords

Data reduction, subset selection, query-aware selection, community-based selection, diversity-aware selection

1 Introduction

The increasing availability of large-scale datasets has made data reduction an important problem in machine learning and data management. The goal is to reduce the size of a dataset while preserving as much information and utility as possible, allowing more efficient storage and processing.

In this work, we study the problem of selecting a representative subset $D' \subset D$ of fixed size B that maximizes a utility function $f(D')$ defined according to the preservation of query answers from the original dataset.

We implement and compare three existing subset selection approaches: (i) exact brute-force optimization, which provides the optimal solution but has limited scalability, (ii) IndepDF, a relevance-based method that assigns individual scores to elements, and (iii) Shapley-value based selection, which estimates the contribution of each element to the objective function.

In addition, we propose a community-based selection strategy that combines IndepDF relevance scores with communities extracted from query co-occurrence patterns. The method allocates the selection budget across communities and optionally applies Maximum Marginal Relevance (MMR) to introduce embedding-space diversity. This formulation allows us to distinguish between two forms of diversity: structural diversity induced by query communities and embedding-space diversity introduced by MMR.

2 Problem Definition

Given a dataset D and a query workload Q, the goal is to select a representative subset

$$D' \subseteq D, \qquad |D'| = B$$

that preserves the query answers of the original dataset as much as possible. The quality of the selected subset is measured through the data forgetting objective:

$$f(D') = \mathbb{E}_{q\sim Q}\left[S(q(D), q(D'))\right] \tag{1}$$

where $q(D)$ represents the answer set of query q over the original dataset, while $q(D')$ represents the answer set obtained when the same query is evaluated over the reduced dataset.

The similarity between the original and reduced query answers is defined as:

$$S(q(D), q(D')) = \frac{1}{|q(D)|} \sum_{d\in q(D)} \max_{d'\in q(D')} \mathrm{sim}(d, d') \tag{2}$$

where $\mathrm{sim}(d, d')$ is the similarity between two data elements.

For the finite query workloads used in our experiments, the expectation can be expanded as:

$$f(D') = \sum_{q\in Q} P(q) \frac{1}{|q(D)|} \sum_{d\in q(D)} \max_{d'\in q(D')} \mathrm{sim}(d, d') \tag{3}$$

where $P(q)$ is the probability associated with query q. In our implementation, queries are explicitly represented as sets of dataset identifiers. Therefore, $q(D)$ corresponds to the elements contained in the query answer set, while $q(D')$ is obtained by restricting the query answer set to the selected subset D'.

3 Methods

3.1 Existing Methods

3.1.1 Method A: Exact Brute-Force Selection. Method A computes the optimal solution for the subset selection problem by performing an exhaustive search over all possible subsets of D with size B.

For every candidate subset:

$$D' \subseteq D, \quad |D'| = B$$

the objective function $f(D')$ is evaluated, and the subset achieving the highest value is returned.

The main advantage of this approach is that it guarantees the globally optimal solution, making it a reference method for evaluating the quality of the other approaches. However, its computational cost grows combinatorially with the size of the dataset. The number of evaluated subsets is:

$$\binom{n}{B}$$

where $n = |D|$. Therefore, brute-force search is only applicable to small datasets.

3.1.2 Method B: IndepDF Relevance-Based Selection. Method B implements IndepDF, a relevance-based selection strategy proposed in [2]. Unlike greedy subset selection approaches, which iteratively optimize the marginal gain of the objective function, IndepDF exploits a specific formulation of the problem based on the Jaccard similarity. Under this assumption, the selection problem can be approximated using independent relevance scores computed from the query workload.

For each element $d \in D$, the relevance score is defined as:

$$w(d) = \sum_{q:d\in q} P(q)\frac{1}{|q|} \tag{4}$$

where $P(q)$ denotes the probability of query q and $|q|$ is the number of elements contained in the query answer set. In our implementation, queries are assumed to be uniformly distributed, i.e.:

$$P(q) = \frac{1}{|Q|}$$

The reduced dataset is obtained by selecting the B elements with the highest relevance scores:

$$D' = \mathrm{Top}_B\{w(d) : d \in D\}$$

This method is computationally efficient because the relevance score of each element is computed independently and the selection process does not require repeated evaluations of the objective function.

The theoretical properties of IndepDF rely on the assumption that the similarity function is the Jaccard similarity. Therefore, the relevance-based decomposition used by IndepDF does not directly generalize to arbitrary similarity measures. Nevertheless, experimental results show that IndepDF can provide effective solutions in practice, while significantly reducing the computational cost compared to approaches based on repeated objective evaluation.

3.1.3 Method C: Shapley-Value-Based Selection. Method C adopts a game-theoretic perspective by estimating the contribution of each element through its Shapley value. The Shapley value measures the average marginal contribution of an element over all possible coalitions of the dataset.

For an element d, the Shapley value is defined as:

$$\phi(d) = \sum_{S\subseteq D\setminus\{d\}} \frac{|S|!(n-|S|-1)!}{n!}\left(f(S\cup\{d\}) - f(S)\right) \tag{5}$$

where $n = |D|$ is the number of elements in the dataset, and S represents a coalition not containing d. The coefficient weights the contribution according to the size of the coalition.

Unlike point-wise relevance methods such as IndepDF, Shapley values evaluate each element across different possible coalitions, allowing interactions among elements to be captured.

The exact computation of the Shapley value for a single element requires evaluating all possible coalitions:

$$O(2^n)$$

and computing the values for all n elements requires:

$$O(n2^n)$$

objective function evaluations. Therefore, exact computation becomes impractical even for moderately sized datasets.

In our implementation, exact Shapley values are used only for small candidate pools, while larger pools can be handled using a Monte Carlo approximation based on random permutation sampling.

Finally, elements are ranked according to their estimated Shapley values and the B highest-scoring elements are selected.

This method provides an importance estimation that accounts for interactions among elements, but it introduces a significant computational cost due to the large number of objective function evaluations required.

3.2 Method D: Query-Aware Community-based Subset Selection (QACSS)

Methods A-C estimate the importance of elements or subsets according to their contribution to the query workload. However, they do not explicitly model structural relationships among elements induced by query co-occurrence patterns.

In many data reduction scenarios, elements are not isolated: their relationships may reveal communities of items that correspond to different regions of the workload. Selecting only the highest-scoring elements may therefore concentrate the budget on a limited portion of the dataset.

To address this limitation, Method D introduces a query-aware community-based selection strategy that combines IndepDF relevance estimation with structural analysis of query co-occurrence patterns.

The method combines three components:

(1) IndepDF-based relevance estimation;
(2) community detection over query co-occurrence relationships;
(3) an optional MMR-based diversification step.

3.2.1 Query-based relevance. The first step estimates the importance of each element using the same IndepDF weighting scheme introduced in Method B. Each query contributes uniformly to the elements appearing in its result set, according to Equation 4. This provides an estimation of how relevant each element is with respect to the workload.

3.2.2 Community discovery and budget allocation. The second component models relationships between elements through a weighted co-occurrence graph. Each node represents an element of D that appears in at least one query result, while an edge connects two elements that appear together in at least one query. The edge weight

corresponds to the number of co-occurrences. The Louvain community detection algorithm is then applied to identify groups of related elements. A fixed random seed is used to ensure that the community assignment is reproducible across executions.

Rather than selecting the highest-scoring elements globally, Method D allocates the selection budget among communities according to their total relevance:

$$B_c = B \frac{\sum_{d \in c} w(d)}{\sum_j \sum_{d \in j} w(d)} \tag{6}$$

where B_c is the budget assigned to community c, B is the total selection budget, and $w(d)$ is the relevance score defined in Equation 4. This encourages selecting representative elements from different communities instead of concentrating the entire budget on the globally highest-scoring elements.

3.2.3 Selection Strategy. After the budget allocation step, elements are selected independently within each community. QACSS supports an optional Maximum Marginal Relevance (MMR) diversification mechanism controlled by the parameter α. When $\alpha = 1$, MMR is disabled and elements are selected exclusively according to their relevance scores. When $0 \leq \alpha < 1$, MMR is activated to balance relevance and embedding-space diversity.

Before applying MMR, the elements in each community are ranked according to their relevance scores, and only the top candidates are retained. Given a community C with allocated budget B_c, the number of candidates considered by MMR is

$$m_c = \min\left(|C|, \max(B_c, \lfloor \gamma B_c \rfloor)\right) \tag{7}$$

where γ is the candidate expansion factor. In our experiments, $\gamma = 2$.

For $\alpha < 1$, the MMR selection score is defined as:

$$score(d) = \alpha R(d) + (1 - \alpha)\left(1 - \max_{s \in S} \text{sim}(d, s)\right) \tag{8}$$

where $R(d)$ is the normalized relevance score obtained by scaling $w(d)$ to the range $[0, 1]$ within the candidate set of the current community. The term $\text{sim}(d, s)$ denotes the cosine similarity between the embeddings of d and s, while S is the set of already selected elements. At each iteration, the candidate with the highest score is selected, and the similarity to the selected set is updated for the remaining candidates.

When $\alpha = 1$, the MMR computation is bypassed and Method D directly selects the B elements with the highest relevance scores within each community. Thus, lower values of α increase the contribution of the diversity term, favoring elements that are less similar to those already selected, while $\alpha = 1$ corresponds to purely relevance-based selection.

The pseudocode of QACSS is provided in Algorithm 1.

3.2.4 Computational complexity. The computational cost of Method D is determined by the three main components of the algorithm. The IndepDF relevance estimation requires a single pass over the query workload, with complexity proportional to the number of query-element occurrences.

The main additional cost introduced by Method D comes from the construction of the co-occurrence graph. For each query, all pairs of returned elements are considered, leading to a cost proportional to

$$O\left(\sum_{q \in Q} |q|^2\right).$$

Therefore, this step becomes more expensive when queries contain a large number of returned elements or when the workload generates a dense co-occurrence graph.

The Louvain community detection algorithm [1] is then applied to the resulting graph and is typically efficient in practice, although its exact complexity depends on the graph structure.

Finally, the MMR selection step requires computing cosine similarities between candidate elements and the already selected elements. Since each embedding has dimensionality k, each similarity computation requires $O(k)$ operations. Given a candidate set of size m and a budget B, the MMR selection step therefore has complexity approximately

$$O(mBk)$$

when similarities are computed on demand.

Therefore, Method D avoids the exponential cost of exhaustive approaches and the repeated objective evaluations required by Shapley-based selection, while introducing an additional overhead due to workload structure analysis and diversity computation.

Algorithm 1 Query-Aware Community-based Subset Selection (QACSS)

Require: Dataset D, query workload Q, embeddings E, budget B, parameter α
Ensure: Selected subset D'

```
Compute IndepDF relevance weights w(d) for d ∈ D
Construct weighted query co-occurrence graph G
Detect communities C in G using Louvain
Allocate budget B across communities according to Σ_{d∈c} w(d)
D' ← ∅
for all c ∈ C do
    B_c ← budget allocated to c
    Rank elements in c by relevance w(d)
    Retain the top m_c = min(|c|, max(B_c, ⌊γ B_c⌋)) candidates
    if α = 1 then
        Select the B_c highest-relevance candidates
    else
        Select B_c candidates using MMR in Eq. 8
    end if
    Add selected elements to D'
end for
if |D'| < B then
    Fill remaining positions with highest-relevance unselected elements
end if
return D'
```

4 Experimental Setup

We evaluate the considered selection methods on both a real-world benchmark dataset and a synthetic dataset under different candidate universe configurations and selection budgets.

All code and experimental configurations used in this work are publicly available at https://github.com/francesco-pittacolo/Data-Reduction-QACSS.

4.1 Datasets

The real-world benchmark dataset used in our experiments is the photo dataset described in [2]. It consists of a collection of photos represented by embedding vectors, together with a query workload where each query is associated with a set of returned photos. The dataset contains 41,620 photos and 443 queries.

To evaluate scalability on a larger dataset, we also generate a synthetic dataset containing 100,000 elements represented by 512-dimensional embedding vectors. The dataset is generated by sampling elements around 100 latent clusters, introducing structured relationships in the embedding space. The query workload consists of 5,000 queries, where each query is generated from a single latent cluster and contains between 20 and 100 elements.

4.2 Evaluation Settings

We consider two different evaluation settings.

First, we evaluate the methods on a restricted candidate pool. The pool contains 18 elements selected using a top-frequency strategy, which retains the elements with the highest occurrence frequency in the query workload. This size was chosen to keep the exhaustive search of Method A and the Shapley-based computation of Method C computationally feasible while still providing a non-trivial selection problem. The top-frequency construction ensures that the candidate universe contains elements that are effectively relevant to the workload. A random selection strategy was not used to construct the pool because it may include elements that never appear in any query or elements with very low frequency, resulting in a less informative candidate universe.

Second, we evaluate scalability on the full dataset, where the candidate universe corresponds to the complete collection of elements. Different selection budgets are considered to analyze how the methods behave as the required subset size changes. Due to the computational cost of the objective evaluation, the full-dataset experiments focus on Method B and Method D. Although Monte Carlo sampling makes Shapley-value estimation feasible for larger candidate pools, the number of objective-function evaluations remains substantial. Therefore, Shapley-based selection was evaluated only on the restricted candidate pool and was not included in the full-dataset experiments.

4.3 Evaluation Metrics

The quality of the selected subsets is evaluated using three metrics. First, the objective value $f(D')$ is used as the primary metric and measures how well the reduced dataset preserves the original query answers. Higher values indicate better preservation of the workload utility.

Second, embedding diversity is measured as the average pairwise distance between the selected elements in the embedding space. The reported values are normalized in the range $[0, 1]$, where larger values indicate more diverse selected subsets.

Finally, execution time measures the computational cost required by each selection method to produce the selected subset.

5 Results

5.1 Quality and Runtime Comparison on the Restricted Candidate Pool

We first evaluate the methods on the restricted candidate pool containing 18 elements. Two different selection budgets are considered, namely $|D'| = 5$ and $|D'| = 7$. This setting allows an exact comparison since Method A can exhaustively evaluate all feasible subsets of the candidate pool and therefore provides the optimal solution. For Method D, we evaluated different values of the MMR parameter α on the restricted candidate pool. Changing α did not produce variations in either the objective value or the measured diversity of the selected subsets. Since the candidate pool is small and contains limited structural variation, the main experiments use $\alpha = 1$. This setting removes the additional similarity computations required by the diversity component of MMR while preserving the same observed objective value and diversity in this scenario. Tables 1 and 2 report the objective values and execution times obtained by the four methods.

Table 1: Objective value on the restricted candidate pool.

	Objective value $f(D')$	
Method	$\|D'\| = 5$	$\|D'\| = 7$
A (Brute Force)	**0.714**	**0.8315**
B (IndepDF)	0.7008	0.8158
C (Shapley)	0.7008	0.8158
D (QACSS)	0.6016	**0.8315**

Table 2: Execution time on the restricted candidate pool.

Method	Time (s)
A (Brute Force)	0.9
B (IndepDF)	**0.0008**
C (Shapley)	36
D (QACSS)	0.003

As expected, Method A provides the optimal objective value for both budgets, since it performs an exhaustive search over all feasible subsets. For a budget of 5 elements, Method B and Method C obtain identical scores, producing solutions that are very close to the optimum. Method D achieves a lower objective value in this configuration.

This behavior can be explained by the characteristics of the restricted candidate pool, which contains the most frequently queried elements due to the top-frequency selection strategy. Under a very limited budget, the community-based allocation may distribute the available selections across different communities instead of concentrating them on the highest-relevance elements, resulting in a lower objective value.

When the budget is increased to 7 elements, Method D reaches the same objective value as the optimal brute-force solution in this restricted setting. This indicates that the performance of the

method is sensitive to the available selection budget, and that a larger budget can improve the quality of the selected subset when community-based selection is applied.

Regarding execution time, the measured values may vary slightly across runs due to execution environment and system conditions. Nevertheless, the results show the expected relative computational cost of the methods. Method C is significantly more expensive than the other approaches due to the repeated objective evaluations required for Shapley value computation. Method A also requires a higher computational cost because it evaluates all feasible subsets through exhaustive search, although its cost remains limited in this small candidate pool.

In contrast, Method B achieves the lowest execution time by computing independent relevance scores, while Method D introduces a small additional overhead due to the construction of the co-occurrence graph and, when the diversity component is active, the similarity computations required by MMR.

5.2 Performance on the Full Dataset

We now evaluate the scalable methods on the complete dataset containing 41,620 elements. Since exhaustive search is computationally infeasible in this setting, only Method B (IndepDF) and Method D (Community-based selection) are considered.

Five selection budgets are evaluated, corresponding to 1%, 2%, 5%, 10% and 20% of the complete dataset, i.e., 416, 832, 2081, 4162 and 8324 selected elements, respectively.

As in the previous experiments, Method D is evaluated with $\alpha = 1$ for the main comparison. In this setting, the MMR diversity component is not activated, and the selection is based only on relevance scores within the detected communities. The effect of different α values is analyzed separately in Section 5.4.1.

Tables 3, 4, and 5 report the objective values, embedding diversity, and execution times obtained on the complete dataset.

Table 3: Objective value $f(D')$ for different selection budgets.

Objective value $f(D')$					
	Budget (% of dataset)				
Method	1%	2%	5%	10%	20%
B	0.4336	0.5789	0.7861	0.8940	0.9756
D ($\alpha = 1$)	**0.5302**	**0.6741**	**0.8172**	**0.8992**	**0.9769**

Table 4: Embedding diversity for different selection budgets.

Embedding diversity					
	Budget (% of dataset)				
Method	1%	2%	5%	10%	20%
B	**0.3933**	**0.3937**	**0.3941**	0.3936	**0.3936**
D ($\alpha = 1$)	0.3932	0.3934	0.3936	**0.3937**	**0.3936**

The results show that Method D consistently achieves higher objective values than Method B across all evaluated selection budgets.

Table 5: Execution time for different selection budgets.

Execution time (seconds)					
	Budget (% of dataset)				
Method	1%	2%	5%	10%	20%
B	**0.007**	**0.008**	**0.008**	**0.008**	**0.012**
D ($\alpha = 1$)	3.891	4.025	4.527	4.640	5.383

The advantage is particularly noticeable for smaller budgets, where the community-based allocation helps preserve relevant elements from different regions of the query co-occurrence structure. As the budget increases, the difference between the two methods becomes progressively smaller, since both methods select a larger fraction of the available elements and therefore converge towards similar solutions.

The embedding diversity values remain very close between the two methods for all evaluated budgets. Since $\alpha = 1$ disables the MMR diversity component, these results indicate that the improvement obtained by Method D is mainly associated with the structural information introduced by the community-based selection strategy.

Regarding execution time, Method B is considerably faster than Method D. Although individual measurements may vary slightly across runs, Method D remains consistently more expensive across all evaluated budgets. This difference is expected since Method B only requires the computation of relevance scores, while Method D introduces the additional cost of constructing the query co-occurrence graph and performing community detection.

5.3 Scalability Evaluation on the Synthetic Dataset

To further evaluate the scalability and structural behavior of the proposed method, we perform additional experiments on a larger synthetic dataset containing 100,000 elements and 5,000 queries. Unlike the real-world benchmark, the synthetic dataset provides a controlled environment in which both the scale of the dataset and the structure of the query workload can be specified. The dataset is generated with a controlled latent structure, where elements are distributed around 100 latent clusters and each query is generated by sampling elements from a single cluster.

This experimental setting primarily allows us to evaluate the scalability of the proposed method on a substantially larger dataset and query workload. At the same time, the latent structure provides an additional controlled source of relationships between elements. Since queries are generated from coherent latent groups, the resulting co-occurrence graph is expected to contain meaningful community structures. Therefore, the synthetic benchmark also provides an opportunity to evaluate whether the community-based selection strategy can exploit relationships induced by the query workload while preserving high query answer similarity.

As in the full-dataset evaluation, exhaustive optimization and Shapley-based selection are not considered due to their computational cost. The comparison is therefore performed between Method B (IndepDF) and Method D (Community-based selection). For consistency with the previous experiments, Method D is evaluated with

$\alpha = 1$, while the effect of different MMR configurations is analyzed separately in Section 5.4.1.

The evaluated selection budgets correspond to 1%, 2%, 5%, 10%, and 20% of the complete dataset, resulting in selected subsets of size 1000, 2000, 5000, 10000, and 20000 elements, respectively.

Tables 6, 7, and 8 report the objective values, embedding diversity, and execution times obtained on the synthetic benchmark.

Table 6: Objective value $f(D')$ on the synthetic dataset

Objective value $f(D')$					
	Budget (% of dataset)				
Method	1%	2%	5%	10%	20%
B	0.6816	0.8577	0.9719	0.9857	**0.9895**
D ($\alpha = 1$)	**0.7955**	**0.9407**	**0.9833**	**0.9864**	**0.9895**

Table 7: Embedding diversity on the synthetic dataset

Embedding diversity					
	Budget (% of dataset)				
Method	1%	2%	5%	10%	20%
B	0.4907	0.4921	0.4933	0.4938	0.4943
D ($\alpha = 1$)	**0.4952**	**0.4949**	**0.4948**	**0.4948**	**0.4947**

Table 8: Execution time on the synthetic dataset

Execution time (seconds)					
	Budget (% of dataset)				
Method	1%	2%	5%	10%	20%
B	**0.088**	**0.156**	**0.145**	**0.145**	**0.102**
D ($\alpha = 1$)	144.937	161.433	159.697	142.258	141.757

The results are consistent with the behavior observed on the real-world benchmark. Method D achieves higher objective values than Method B for the evaluated budgets up to 10%, while the two methods obtain the same objective value at the 20% budget. The largest improvements are obtained for small budgets. As the selection budget increases, the gap between the two methods gradually decreases, indicating that the advantage of incorporating workload structure is more relevant in highly constrained selection scenarios.

The embedding diversity values, with $\alpha = 1$, are similar between the two methods for all evaluated budgets, with slightly higher values for Method D.

Regarding execution time, Method D introduces a substantial computational overhead compared with IndepDF. This difference is mainly caused by the construction of the co-occurrence graph and the subsequent community detection step. In contrast, IndepDF only requires the computation of independent relevance scores and therefore remains significantly faster.

Overall, these results highlight the trade-off introduced by the proposed method: community-based selection improves subset quality by exploiting structural information from the workload, but requires additional computation compared with purely relevance-based selection.

5.4 Effect of the MMR Parameter

5.4.1 Real-World Dataset. Finally, we analyze the influence of the MMR parameter α on Method D. The parameter is varied while keeping all remaining configurations fixed. To evaluate whether the effect of MMR changes with the available selection budget, the analysis is performed for three representative budgets: 1%, 5%, and 20% of the complete dataset.

Tables 9, 10, and 11 report the objective value and embedding diversity obtained for different values of α.

Table 9: Effect of α with a 1% selection budget (416 elements)

Budget: 1% of dataset		
α	$f(D')$	Embedding diversity
0.00	**0.5315**	**0.4037**
0.25	0.5224	0.4007
0.50	0.5268	0.3990
0.75	0.5241	0.3973
1.00	0.5247	0.3930

Table 10: Effect of α with a 5% selection budget (2081 elements)

Budget: 5% of dataset		
α	$f(D')$	Embedding diversity
0.00	0.7898	**0.4041**
0.25	0.8027	0.4025
0.50	0.8113	0.4006
0.75	0.8153	0.3985
1.00	**0.8176**	0.3935

Table 11: Effect of α with a 20% selection budget (8324 elements)

Budget: 20% of dataset		
α	$f(D')$	Embedding diversity
0.00	0.9713	**0.3973**
0.25	0.9749	0.3967
0.50	0.9762	0.3959
0.75	0.9768	0.3951
1.00	**0.9769**	0.3936

The results show that the influence of the MMR parameter depends on the available selection budget. For the 1% budget, different

values of α produce only limited variations in the objective value. In this setting, the highest objective value is obtained for $\alpha = 0$, while increasing α mainly reduces the embedding diversity. This suggests that when the selection budget is extremely limited, preserving diversity can also contribute to maintaining representative elements in the selected subset.

As the selection budget increases, the expected relevance-diversity trade-off becomes progressively more evident. For the 5% budget, increasing α produces a consistent improvement in the objective value, reaching the best result for $\alpha = 1$, while the embedding diversity decreases gradually. This indicates that a stronger focus on relevance allows the method to select elements that better optimize the objective, at the cost of a moderate reduction in diversity.

The same behavior is observed for the 20% budget, although the differences between the values of α become smaller. Since a larger fraction of the dataset is selected, all configurations tend to converge toward similar solutions, reducing the impact of the diversity component on the final selection.

Overall, these results confirm that the MMR parameter effectively controls the balance between relevance and diversity. Lower values of α favor more diverse subsets, whereas larger values prioritize relevance and improve the objective value. Since $\alpha = 1$ consistently achieves the highest or near-highest objective values across the evaluated budgets, this configuration is adopted for the main comparison with the other selection methods.

5.4.2 Synthetic Dataset. To further evaluate the generality of these observations, we repeat the MMR sensitivity analysis on the synthetic dataset using the same relative selection budgets. Since the synthetic dataset contains 100,000 elements, the 1%, 5%, and 20% budgets correspond to 1,000, 5,000, and 20,000 selected elements, respectively. Using the same relative budgets allows a direct comparison between the real-world and synthetic datasets.

Tables 12, 13, and 14 report the objective value and embedding diversity for the different values of α.

Table 12: Effect of α with a 1% selection budget

Budget: 1% of dataset (1,000 elements)		
α	$f(D')$	Embedding diversity
0.00	0.7797	0.4951
0.25	0.7951	**0.4952**
0.50	0.7953	**0.4952**
0.75	**0.7955**	**0.4952**
1.00	**0.7955**	**0.4952**

The synthetic dataset exhibits a different behavior from the real-world benchmark. For the 1% budget, increasing α from 0 to 0.25 produces a substantial improvement in the objective value, from 0.7797 to 0.7951. Further increases in α result only in marginal changes, with the objective reaching 0.7955 for $\alpha = 0.75$ and $\alpha = 1$. A similar pattern is observed for the 5% and 20% budgets: the objective improves when moving from $\alpha = 0$ to $\alpha = 0.25$, after which the differences become negligible.

In contrast to the real-world dataset, embedding diversity remains almost constant across all α values, with values around 0.495

Table 13: Effect of α with a 5% selection budget

Budget: 5% of dataset (5,000 elements)		
α	$f(D')$	Embedding diversity
0.00	0.9813	0.4948
0.25	**0.9833**	0.4948
0.50	**0.9833**	0.4948
0.75	**0.9833**	0.4948
1.00	**0.9833**	0.4948

Table 14: Effect of α with a 20% selection budget

Budget: 20% of dataset (20,000 elements)		
α	$f(D')$	Embedding diversity
0.00	0.9880	0.4947
0.25	**0.9895**	0.4947
0.50	**0.9895**	0.4947
0.75	**0.9895**	0.4947
1.00	**0.9895**	0.4947

for all three budgets. This suggests that, for the synthetic dataset, the MMR parameter has a limited effect on the measured diversity of the selected subsets. The strong cluster structure introduced during synthetic data generation may make the selected subsets relatively stable in terms of embedding diversity even when the relevance-diversity weighting is changed.

Overall, the two datasets show that the effect of α is dataset-dependent. On the real-world dataset, increasing α produces a clearer relevance-diversity trade-off, particularly for larger selection budgets. On the synthetic dataset, the main improvement occurs when moving from the purely diversity-driven configuration ($\alpha = 0$) to a configuration that incorporates relevance ($\alpha = 0.25$), while further increases in α provide little additional benefit. Nevertheless, $\alpha = 1$ remains a strong configuration on both datasets and is therefore used in the main comparison experiments.

6 Discussion

Taken together, the experimental results provide a consistent picture of when community-based selection is effective and when its benefits are more limited.

Conditions under which Method D is effective. Across the full real-world and synthetic datasets, Method D achieved higher or equal objective values than IndepDF at every tested budget, with the largest improvements occurring at the smallest budgets (1–2% of the dataset). This behavior is consistent with the motivation behind the proposed method: when the selection budget is scarce, allocating selections across different communities can prevent IndepDF from concentrating the budget on a limited region of the query workload. By distributing selections according to Equation 6, Method D can preserve relevant elements from multiple regions of the workload that a purely relevance-based ranking may overlook.

Conditions under which Method D is less effective. The restricted-pool experiment (Table 1, $|D'| = 5$) shows the opposite

behavior. When the budget is small in absolute terms and comparable to or smaller than the number of detected communities, the allocation mechanism may distribute selections too thinly across communities. In this setting, Method D performs worse than both IndepDF and Shapley. These results are therefore not contradictory: they indicate that the effectiveness of community-based allocation depends not only on the absolute selection budget, but also on the relationship between the budget and the number of communities.

Source of the observed improvement. Embedding diversity was nearly identical between Method B and Method D at $\alpha = 1$ in both the full-dataset and synthetic experiments (Tables 4 and 7), even though Method D achieved higher objective values. This indicates that the improvement is primarily attributable to the structural diversity introduced by community-aware allocation rather than to embedding-space diversity, which is explicitly introduced only when $\alpha < 1$. The MMR analysis further supports this interpretation. On the real-world dataset, $\alpha = 1$ (no MMR) achieved the best or near-best objective value across all evaluated budgets, suggesting that the community-allocation mechanism alone accounts for most of the observed improvement over IndepDF.

Computational cost. The improvements achieved by Method D come at a substantial computational cost. On the full real-world dataset, Method D was roughly 400–500× slower than IndepDF, while on the synthetic dataset it was more than 1000× slower. This overhead is primarily caused by co-occurrence graph construction and Louvain community detection rather than by the final selection step. The practical impact of this cost therefore depends on the intended use case: for one-time or infrequently repeated offline data reduction, the additional computation may be acceptable, whereas frequent recomputation as the workload evolves may favor the lower cost of relevance-based methods such as IndepDF.

Dataset- and budget-dependence of MMR. The MMR analysis shows that the relevance-diversity trade-off is not universal across datasets and selection budgets. On the real-world dataset, increasing α produces a relatively smooth trade-off between objective value and embedding diversity, particularly at larger budgets. In contrast, on the synthetic dataset, most of the improvement obtained by moving away from pure diversity ($\alpha = 0$) occurs already at $\alpha = 0.25$, after which the objective remains nearly unchanged and embedding diversity is almost constant. These differences suggest that the effect of MMR depends on the structure of the underlying workload and embedding space. Consequently, the parameter α should be tuned according to the characteristics of the dataset and the available selection budget rather than fixed a priori.

7 Conclusion

This work investigated subset selection strategies for data reduction, with a particular focus on scalable selection under large query workloads. We compared exact optimization, IndepDF, Shapley-value-based selection, and the proposed community-based selection strategy, which combines relevance-based scoring with structural information extracted from query co-occurrence patterns.

The experimental results show that the proposed method can improve query answer preservation compared with IndepDF, particularly when the available selection budget is small. The results on both the real-world and synthetic datasets suggest that exploiting workload structure can help avoid over-concentration on a limited portion of the dataset and preserve representative elements from different regions of the query workload. The synthetic experiments further demonstrate that this approach remains applicable at a substantially larger scale.

At the same time, the results highlight an important trade-off. The community-based approach introduces a substantial computational overhead due to the construction and analysis of the query co-occurrence graph. Therefore, its main advantage is not lower computational cost, but improved subset quality when the selection budget is constrained. This makes the method more suitable for offline or infrequently repeated data reduction scenarios, where the additional preprocessing cost can be justified by the quality of the resulting subset.

Finally, the MMR experiments show that embedding-space diversity provides additional benefits that depend on both the dataset and the selection budget. While community-based structural diversity already provides a substantial improvement over purely relevance-based selection, the relevance-diversity trade-off introduced by MMR is dataset-dependent. Future work could therefore investigate more adaptive community allocation and MMR strategies, as well as methods for reducing the computational cost of co-occurrence graph construction and community detection on even larger datasets.

References

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